Work Like a Physicist

Student workbook: answers

Model answers for every question, table and checkpoint in the workbook. Where a task is practical or personal (your own results, your own graph) the entry gives sample data, a worked example or the marking points to look for. Accept any answer that makes the same point in other words.

Lesson 1 · Can this data be trusted? — WORKBOOK PP. 4–8

Task 1a · p. 4

Set A. Its readings (1.20, 1.23, 1.19) are close together, so a single reading from A is likely to be near the mean. B's readings run from 1.05 to 1.47 s, so any one of them could be a long way from the truth, even though the mean happens to be the same.

Task 1b · p. 5

C's repeats are tightly clustered (precise), but that says nothing about whether they are correct. C's mean (0.91 s) is 0.30 s lower than A and B for the same car and ramp, so a **systematic error** is likely, such as the stopwatch being stopped too early every time or the car starting from a different point. Precise but not accurate.

Task 1c · p. 5

Repeats show how much the readings vary and so how far to trust the data. They allow a mean to be calculated, which reduces the effect of random error, and a single odd value can be spotted.

Task 2 · p. 5

Situation	Example from a school lab (accept any sensible one)
Accurate and precise	Three timings of a pendulum swing of 1.50, 1.51, 1.50 s when the true value is 1.50 s.
Precise but not accurate	Measuring with a ruler whose end is worn off: lengths of 14.2, 14.2, 14.3 cm when the true length is 15.0 cm.
Accurate on average, not precise	Hand-timed drops of 0.9, 1.4, 1.1 s: widely spread, but the mean is close to the true 1.1 s.
Neither	A balance that was not zeroed, used on a wobbly bench: 52, 58, 49 g for a 50 g mass.

Task 2a · p. 5

Precise but not accurate. The results were consistent (precise) but all wrong by the same amount, because of the loose cable. Everything shifted the same way, so it was a systematic error.

Task 4 · p. 6

Set	Mean	Range	Uncertainty
A	12.2 (12.17)	12.0 to 12.4	0.2
B	8.7	8.4 to 9.1	0.35 (≈ 0.4)
C	25.0 (25.03)	24.9 to 25.1	0.1
D	3.37 (3.4)	3.20 to 3.65	0.225 (≈ 0.23)

Range = the maximum and minimum values. Uncertainty = (largest − smallest) ÷ 2. Accept one decimal place more or less than shown, if used sensibly.

Task 4a · p. 6

Set A: (12.0 + 12.4 + 12.1) ÷ 3 = 36.5 ÷ 3 = 12.17. (Set B: 26.1 ÷ 3 = 8.7. Set C: 75.1 ÷ 3 = 25.03. Set D: 10.10 ÷ 3 = 3.37.)

Task 4b · p. 6

Most precise: **C** (smallest spread: its readings differ by only 0.2). Largest uncertainty: **B** (0.35).

Task 4c · p. 6

D: 3.65 is far from 3.20 and 3.25, so it may be a mistake (a possible anomalous result). It should be repeated before the mean is trusted. Accept B (largest spread) if justified.

Task 5a · p. 6

A systematic error moves every reading the same way (for example all too high). Averaging many repeats just gives a mean that is wrong by the same amount, so repeating does not fix it. Random errors go up and down, so they cancel out in the mean.

Task 5 · R/S · p. 6

Problem	R/S
Stopwatch started slightly late each time	S (arguable: S if the delay is always about the same; R if it varies from trial to trial). Either with a reason is fine.
Judging bounce height by eye	R
A draught from an open window	R
Ruler starts measuring from its very end	S
Balance reads 0.2 g with nothing on it	S (zero error)
Reading a scale at an angle, always the same side	S (parallax)

Task 6 · p. 7

Instrument	Resolution	Could it detect 0.5 mm?
Ruler marked every 1 cm	1 cm (10 mm)	No
Ruler marked every 1 mm	1 mm	No (the change is smaller than the smallest division)
Stopwatch to 0.01 s	0.01 s	not a length
Balance to 0.1 g	0.1 g	not a length

Task 7 · p. 7

Personal results. Look for: three readings recorded with units; mean = total ÷ 3; uncertainty = (largest − smallest) ÷ 2. **Sample (dropped-ruler reaction test):** catch distances 14.0, 17.5, 15.5 cm → mean 15.7 cm, uncertainty (17.5 − 14.0) ÷ 2 = 1.75 ≈ 1.8 cm.

Task 7a · p. 7

e.g. a 30 cm ruler with a resolution of 1 mm (0.1 cm).

Task 7b · p. 7

e.g. reaction time: sometimes the ruler was caught quickly and sometimes late. The readings varied up and down (14.0, 17.5, 15.5 cm) with no pattern, which is what random error does.

Task 7c · p. 7

e.g. the zero of the ruler was not level with the fingers each time, or the same person always started the stopwatch late. To find it: check the instrument against another one, check the zero, change the person or method and see whether the whole set shifts.

Task 8 · p. 8

Claim	What I need to know first	Suspected error
"Reaction times improved after training." Two tests were timed by different people.	Were the timers the same person? How big is the difference compared with the spread of repeats? How many repeats were taken?	Systematic (a different timer has a different start/stop delay, shifting every reading)
"This spring stretches 4.0 cm per 100 g." One reading each.	Were repeats taken? What is the uncertainty? Does a single value represent a pattern across several masses?	Random (a single reading cannot show or reduce it)
"The room is exactly 6.000 m long." One reading, slack tape.	How many repeats? Was the tape tight and straight? What is the tape's resolution (the third decimal place is not credible)?	Systematic (slack tape always reads too long)

Checkpoint · p. 8

1. precise.

2. accurate.

3. the largest and smallest repeat readings: (largest − smallest) ÷ 2.

4. random … systematic, because a systematic error shifts every reading the same way, so averaging repeats leaves the mean just as wrong.

5. Open. e.g. take more repeats, use an instrument with finer resolution, use the same timer each time, or check the instrument's zero.

Take it home · p. 8

Open. A good question names something that is missing: how many people/trials the claim came from, how it was measured, or what the comparison was.

Lesson 2 · What are you actually changing?

WORKBOOK PP. 9–11

Task 1 · p. 9

Investigation	Independent	Dependent	Two controls (examples)
Ramp height and trolley speed	Ramp height (cm)	Speed of trolley (or time over a fixed distance)	Same trolley/mass; same release point on the ramp; same surface
Best thermal insulator	Insulating material	Final temperature (or temperature fall in a set time)	Same starting temperature; same volume of water; same time; same beaker
Elastic bands and launch distance	Number of elastic bands	Launch distance (cm)	Same projectile; same pull-back; same launch angle
Which drink has the most sugar	Type of drink	Amount of sugar (e.g. mass of sugar or sweetness reading)	Same volume of drink; same temperature; same test method

Task 1a · p. 9

Open. e.g. ramp height: if the trolley is released from a different point or has a different mass each time, a longer distance could be caused by those changes and not by the ramp height, so the conclusion would not be valid. The result would not be a fair test.

Task 3 · p. 10

Investigation	Graph	Independent variable
Temperature and dissolving time	LINE	temperature: continuous
Surface with most friction	BAR	type of surface: categoric
Mass and stretch of a spring	LINE	mass: continuous
Paper towel that absorbs the most water	BAR	type/brand of paper towel: categoric
Pendulum length and time period	LINE	length: continuous
Best sound-absorbing material	BAR	type of material: categoric

Task 4 · p. 10

Independent variable: **surface type** (bench, carpet, rubber mat, sandpaper). Dependent variable: **distance the car travels** (cm). Graph type: **bar chart**, because surface is categoric.

Task 4a · p. 11

Same car, same ramp and ramp height, same release point (and release it without a push), same surface after the ramp where not being tested. The hardest to keep the same is usually the release (starting without a push, in the same position each time) *or* the length of the surface being tested. Accept any one starred with a reason.

Task 5a · p. 11

Any two of: the balls are dropped from different heights (window v. hand), so it is not a fair test; the release method differs; only one ball of each kind and no repeats; “better” is not defined; “expensive” is not a measurable property of the ball.

Task 5b · p. 11

e.g. “How does the type of tennis ball affect its bounce height when dropped from 1.0 m?”

Task 5c · p. 11

A **bar chart**, because the independent variable (type of ball) is categoric.

Checkpoint · p. 11

1. change.

2. measure.

3. the independent variable is continuous (a number on a scale, such as time, length or temperature).

4. the independent variable is categoric (names or groups, such as material or colour).

5. The graph shows what happens when the thing you changed changes. If what you changed is a number on a scale, points can sit along a line; if it is separate groups, there is nothing between the groups to join with a line. The dependent variable is nearly always a number, so it cannot decide.

Take it home · p. 11

Open. Check each question names one thing changed and one thing measured, and that the independent variable is correctly labelled continuous or categoric (at least two bar-chart and two line-graph questions).

Lesson 3 · The drop test

WORKBOOK PP. 12–15

Task 1 · p. 12

Independent variable: **material**. Dependent variable: **bounce height / cm**. Controls: same ball; same drop height; same way of releasing; same surface under the material; same person reading the height. Categoric. A **bar chart**, because the material is categoric.

Think first · p. 13

The **lowest** bounce. A good shock absorber soaks up energy instead of returning it, so less energy is left to make the ball bounce.

Task 2 · p. 13

Practical. Any agreed drop height that is the same every time (e.g. 100 cm). Safety: drop vertically only.

Task 3 · p. 13

Personal results. **Sample results (drop height 100 cm; yours will differ):**

Material	Bounce 1 / cm	Bounce 2 / cm	Bounce 3 / cm	Mean / cm
Sponge	18	21	18	19.0
Bubble wrap	23	20	23	22.0
Cloth	25	25	22	24.0
Cardboard	30	33	33	32.0
Rubber mat	38	41	39	39.3

Task 3a · p. 13

e.g. sponge: (18 + 21 + 18) ÷ 3 = 57 ÷ 3 = 19.0 cm.

Task 3b · p. 13

Open. A good answer names the reading, says it was repeated or checked, and does not delete it silently, e.g. “bounce 2 for cardboard looked high, so I repeated it and recorded both.”

Task 4 · p. 14

Check against the workbook checklist: title; material on the x-axis; y-axis “Mean bounce height / cm”; scale in sensible steps filling the grid; bars of equal width with gaps; bar heights equal to the **means** (not trial 1). Order of bars does not matter, because the categories have no order.

Task 5a · p. 15

The material with the **lowest** mean bounce. Sample: sponge, mean bounce 19.0 cm compared with 39.3 cm for the rubber mat.

Task 5b · p. 15

Repeats show how much the readings vary, allow a mean to be calculated (which reduces random error), and let you spot an odd result. One drop might be a fluke.

Task 5 · errors · p. 15

Error type	What went wrong (examples)	How to reduce it
Random	Judging the top of a fast bounce by eye; releasing slightly differently each time	Take more repeats; record with a phone video in slow motion; use a clamp or guide to release the ball
Systematic	Always reading the ruler from above (parallax); ruler not vertical; ball always released slightly above the marked height	Read at eye level against a vertical ruler (set square); mark the drop height; check the zero

Task 5c · p. 15

No. A systematic error shifts all the readings the same way, so more repeats just give a mean that is just as wrong. The cause has to be found and fixed (e.g. use a set square to make the ruler vertical).

Task 6a · p. 15

Any two of: a different ball or drop height; different thickness or sample of each material; a different surface under the materials; reading the bounce height in a different way; different resolution of ruler. No-one need be wrong: the practical was not controlled in the same way.

Task 6b · p. 15

A **line graph**: thickness is continuous, so the thickness (cm) goes on the x-axis and mean bounce height on the y-axis, with a best-fit line. The independent variable has changed from categoric to continuous.

Checkpoint · p. 15

1. categoric (the type of material).

2. (sample) sponge … lowest.

3. e.g. judging the bounce height by eye, or releasing the ball differently each time.

4. Open. e.g. use slow-motion video to read the bounce, or clamp the ball release.

Lesson 4 · Higher ramp, further flight?

Task 1 · p. 16

Independent variable: **ramp height** (cm). Dependent variable: **distance travelled** (cm). Continuous. A **line graph**. Four controls: same trolley (mass); same release point on the ramp; same ramp length/surface; same floor surface. Also no push at release.

Task 3 · p. 17

Personal results. **Sample results (yours will differ):**

Ramp height / cm	Distance 1 / cm	Distance 2 / cm	Distance 3 / cm	Mean / cm
5	21	24	24	23.0
10	41	44	41	42.0
15	60	63	60	61.0
20	78	81	81	80.0
25	100	100	97	99.0

Task 3a · p. 17

From the sample: largest mean 99.0 cm; smallest mean 23.0 cm.

Task 3b · p. 17

The axes must fit the data: the scale has to run from below the smallest to just above the largest value, so the points fill the grid instead of being squashed into a corner.

Task 4 · p. 18

Example from the sample, if the grid is 10 big squares high: y from 0 to 100 cm with each big square worth 10 cm; x from 0 to 25 cm with each big square worth 5 cm (or 2 cm, if there are enough squares). Check that the largest value fits, the steps are 1, 2, 5, 10, 20 or 50 per square, and that the data fills most of the grid.

Task 5 · p. 18

Practical. Ramp height on the x-axis, mean distance on the y-axis, points as small crosses, both axes labelled with units, and **no line yet** (the best-fit line is added in Lesson 5).

Task 6 · p. 19

Self- and peer-assessment against the six listed mistakes. Check that no mistakes remain.

Checkpoint · p. 19

1. ramp height … centimetres (cm).

2. distance travelled … centimetres (cm).

3. ramp height is continuous (it can take any value on a scale).

4. Open. e.g. the release (starting from the same point without a push), because the trolley was easy to give a small push or start from a slightly different spot.

5. Own answer from own points. With the sample data: a straight line, because the mean distance rises by roughly the same amount (about 19 cm) for every 5 cm of ramp height.

Lesson 5 · The point that does not fit

WORKBOOK PP. 20–24

Task 1a · p. 20

The best-fit line (or curve) shows the pattern most clearly, because it shows the overall trend and is not thrown off by each small error.

Task 1b · p. 20

Dot-to-dot claims that every measurement is exactly right and that the quantity zig-zags between points. That is not true: every measurement has some uncertainty (Lesson 1), so the points scatter around the true pattern.

Task 2a · p. 21

That the measurements are never perfect: they contain random errors, so points do not lie exactly on the trend. The line shows the pattern underneath the scatter.

Task 3 · p. 21

Plot (5, 20), (10, 37), (15, 52), (20, 96), (25, 80). The best-fit line through the trend passes close to (5, 20), (10, 37), (15, 52) and (25, 80): roughly distance = 3.1 × height + 5, giving about 67 cm at 20 cm, and has the (20, 96) point clearly off the line.

Task 3a · p. 21

96 cm at 20 cm. It is about 30 cm above where the trend suggests (around 67 cm), and it is larger than the value at 25 cm, so the distance seems to fall as the ramp gets higher.

Task 3b · p. 22

Not delete it. First **check**: was it recorded or copied correctly (e.g. 66 or 69 written as 96)? Was the method the same? Repeat the measurement at 20 cm if possible. Only if it is explained as a mistake should it be corrected or repeated.

Task 3c · p. 22

Any two: trolley pushed or released from a higher point on the ramp; the ramp was set to the wrong height; distance misread or recorded wrongly (e.g. 66 as 96); a different surface/ trolley.

Task 4 · p. 22

Personal graph. A smooth best-fit line (ruler if straight), balanced points either side, anomalous results circled with a note, or "no anomalous results", plus all the checklist items.

Task 5 · p. 22

Peer review. Look for a specific improvement, e.g. "your scale on the x-axis goes up in 3s, use 5s" or "label the y-axis with the unit", not just "be neater".

Task 6 · p. 23

Model answer (using the sample ramp data): As the ramp height increased, the distance travelled by the trolley *increased*. I know this because the graph shows *a line going steadily upwards*. For example, at *5* cm the mean distance was *23* cm, while at *25* cm it was *99* cm. One possible source of random error was *judging where the trolley stopped by eye*. One result that may be an anomalous result is *(none)* because *all points lie close to the line*. Evidence must be quoted from the student's own data.

Task 7a · p. 23

Axes not labelled

No units

Points joined dot-to-dot (not a best-fit line)

All the data squashed into the bottom-left corner (poor scale)

Task 7b · p. 23

Accept either of: missing axis labels and units, because without them nobody can tell what the graph shows, or a poor scale, because the pattern cannot be read. Missing labels/units is the best answer.

Task 7c · p. 23

Mass / g on the x-axis (0 to 500 g, e.g. 100 g per 2 big squares), extension / cm on the y-axis (0 to 12 cm, 2 cm per big square), points plotted as small crosses at (100, 2.1), (200, 4.0), (300, 6.2), (400, 7.9), (500, 10.1), a ruled straight best-fit line through the origin (about 2.0 cm per 100 g), with a title and labelled axes.

Checkpoint · p. 24

1. a single smooth line (straight or curved) that shows the overall trend, with points balanced either side.

2. it claims every reading is exactly right and ignores the uncertainty in measurements.

3. a result that does not fit the pattern of the others.

4. check how it was recorded and measured, and repeat the measurement if possible.

5. Removing a result just because it does not suit what you expected hides evidence and gives a false picture. It misleads the reader, which is a breach of honesty, not just a messy graph.

Take it home · p. 24

Open. Check that the student has run the graph against the quality checklist: both good points are specific (e.g. “clear title”) and the weakness would genuinely lose marks (e.g. no units, a truncated or inconsistent scale).

Lesson 6 · Solo flight

WORKBOOK PP. 25–29

Task 1 · p. 26

Personal planning. **Model answer:** How does wing length affect the fall time of a paper helicopter? Independent: wing length / cm. Dependent: fall time / s. Controls: same paper; same number of paperclips; same drop height; same person timing; same wing width. Continuous, so a line graph. Five values: 4, 5, 6, 7, 8 cm. Safety: no standing on chairs or benches; use a height reachable from the floor.

Task 2 · p. 27

A table with a column for wing length / cm, three trial columns (Time 1 / s, Time 2 / s, Time 3 / s), and a Mean time / s column. Units go in every heading and a row for each of at least five wing lengths. **Sample results (yours will differ):**

Wing length / cm	Time 1 / s	Time 2 / s	Time 3 / s	Mean time / s
4	1.1	1.2	1.1	1.13
5	1.4	1.5	1.4	1.43
6	1.7	1.8	1.7	1.73
7	2.1	2.0	2.1	2.07
8	2.4	2.5	2.4	2.43

Task 3a · p. 27

Open. A good answer names the trial and the reason (e.g. "wing length 6 cm, trial 2: helicopter hit the bench") and says it was repeated and the original noted.

Task 4 · p. 28

Personal graph: wing length on x-axis, mean fall time on y-axis, small crosses, labelled axes with units, a sensible scale and a best-fit line (a smooth straight or curved line, not dot-to-dot). The sample data give a rising line, roughly straight.

Task 5 · p. 28

Model answer (sample data): As *wing length* increased, the fall time *increased*. The evidence for this is *the best-fit line sloping upwards steadily with all points close to it*. At *4 cm* the mean was *1.13 s*, at *8 cm* it was *2.43 s*. My graph *does not* contain a possible anomalous result (all points lie close to the line).

Task 5a · p. 28

Read from the student's best-fit line at a length that was not tested but lies inside the range tested. **Sample:** at 5.5 cm the line gives about 1.6 s (read upwards from 5.5 cm on the x-axis to the line, then across).

Task 6 · p. 29

Self-assessment against the nine success criteria; there is nothing to mark apart from checking that each box is honestly ticked. Use the marking points in Tasks 1–5.

Checkpoint · p. 29

Open. Strong answers are specific, e.g. "scale in even steps that fill the grid"; "record each reading immediately and take three trials"; "keep the same person timing".

Take it home · p. 29

Model answers. Strength: three trials at every wing length with the same timer. Weakness: only five wing lengths. Random error: reaction time starting and stopping the stopwatch. Improvement: use a phone video or a second timer and more repeats.

Lessons 7–8 · Your investigation

WORKBOOK PP. 30–38

Task 2 · p. 31 (rejected questions)

Rejected question	What is wrong
"What is the best paper helicopter?"	Too vague: "best" is not defined and no variable is named, so there is nothing to change or measure.
"How does height affect it?"	"It" is not defined; the dependent variable is unclear (what is measured?), and so is which height.
"How does surface affect speed and distance and time?"	Three dependent variables. A good investigation measures one.
"How does throwing force affect distance?"	Force cannot be measured or kept constant fairly by hand; the independent variable can't be controlled (and is difficult to do safely).

Task 3 · p. 32

Personal planning. Approve against the teacher-checkpoint criteria: a testable "How does X affect Y?" question; one independent variable with unit; one measurable dependent variable with unit; at least three realistic control variables; continuous or categoric correctly stated; the matching graph with a reason; 4–5 values; safety risks with controls.

Task 4 · p. 32

Numbered steps that someone else could follow: equipment set-up; how the independent variable is changed; how the dependent variable is measured; at least three trials per value; how each control variable is kept the same; what is done with the results. Safety steps count.

Task 5 · p. 33

Teacher sign-off: use the eight criteria in the box.

Task 6 · p. 33

Table with the independent variable (with unit) as the first heading, three trial columns, a mean column, units in headings, and four or five rows.

Task 6a · p. 33

Mean = total of the three trials ÷ 3. For example (12.2 + 12.4 + 11.8) ÷ 3 = 36.4 ÷ 3 = 12.1.

Task 6b (lab notes) · p. 34

Lab notes: anything that happened (equipment problems, trials repeated, a control that slipped). It should be written during the work, not later.

Checkpoint · p. 34

Personal. Look for a full question in the form "How does X affect Y?"; independent variable with a continuous/categoric label; a dependent variable that can be measured; a control actually kept constant; a result flagged to check.

Task 7 · p. 35

Continuous independent variable → line graph (x-axis: independent variable; y-axis: mean of the dependent variable; units on both axes). Categoric → bar chart (x-axis: categories; y-axis: mean of the dependent variable with unit). Largest mean value is used to choose a scale: divide by the number of big squares and round *up* to 1, 2, 5, 10, 20 or 50 per square.

Task 8 · p. 35

Personal graph: use the Lesson 3–5 checklists (title; labelled axes with units; sensible scale; accurate crosses or equal-width bars with gaps; best-fit line for a line graph; anomalous results circled).

Task 9 · p. 36

Personal. Strong answers describe the pattern in words ("as X increased, Y increased"), judge whether it is clear or scattered, name any anomalous result or say there are none, and quote the single strongest number.

Task 10 · p. 36	Conclusion marking points: states the relationship in words; quotes **two data values** from the group's own results ("at 5 cm the mean was …, at 25 cm it was …"); (line graph) uses "as … increased, … increased/decreased"; (bar chart) names the category with the highest or lowest mean; links back to the question ("This suggests that …").
Task 11 · p. 36	Evaluation marking points: a specific **strength** (e.g. three trials, same timer); a specific **weakness** (e.g. only four values); a **random** error named with its cause (e.g. reaction time on the stopwatch); a **systematic** error named with its effect (e.g. ruler not zeroed, always reading too high); an improvement that addresses the weakness (e.g. use a light gate, test more values). "Human error" on its own earns nothing.
Task 11 (confidence) · p. 37	Open. Confidence should be tied to the evidence: how clear the pattern is, how big the spread of repeats is, how many values were tested. Ways to increase confidence: more repeats, more values, a more precise instrument, repeating with a different group or method.
Task 12 · p. 37	Peer review: tick each of the sixteen criteria honestly. "Strongest part" and "question" should be specific.
Task 13a · p. 37	Open. A good answer compares numbers ("we both found it doubled between …") and explains any difference (different equipment, control variable, resolution, systematic error).
Task 13b · p. 37	Linear if the points follow a straight best-fit line (the dependent variable changes by the same amount for each equal step of the independent variable); non-linear if the best-fit line curves. A straight line through the origin shows direct proportionality.
Task 13c · p. 37	Open. Should justify a range: e.g. extend beyond the tested range to see whether the pattern continues, or test more values between the existing ones to see whether the pattern is straight or curved.
Checkpoint · p. 38	Personal; no single correct answer. The strongest answers refer to specifics: a clear decision (e.g. "controlling the release point"), a number from their results, and a named source of error (not "human error").
Take it home · p. 38	Open, individual reflection. Q4 expects a realistic objection a stranger could make, such as "three trials is not many", "you only tested five values", or "how do you know the timer was accurate?".